Scalability & Future Plan

Date	29 september 2026
Team ID	6ab2ac2a1266c775575a9e56
Project Name	Farm Management System
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state, ensuring long-term viability and impact.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited number of users can be handled initially	More users may affect system performance	High
2	Limited data storage and reporting features	Large farm records may become difficult to manage	Medium
3	Basic security and notification features	User data protection and timely alerts can be improved	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	System is designed for a limited number of users.	Use a scalable server and database to support more farmers and users simultaneously.

2	Data Storage	Farm, crop and activity data are stored in a structured database.	Increase database capacity and use cloud storage for handling large amounts of farm data.

3	Performance	System works efficiently with the current number of users and records.	Optimize database queries and server performance to support more users and data.
4	Security	Basic login and user authentication are provided.	Implement stronger authentication, password encryption, role-based access and regular data backup.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Mobile application and notification system	3–6 Months	Farmers can access the system easily and receive timely alerts.
Phase 3	Weather integration and smart irrigation support	6–12 Months	Helps farmers make better decisions about irrigation and crop activities.
Phase 4	AI-based crop recommendations and analytics	12–18 Months	Provides useful insights for crop planning and improves farm management.